

EXECUTIVE SUMMARY

This project analyzes YouTube Trending Videos data across 10 countries to identify the factors influencing video popularity, engagement, and trending performance. The dataset consists of over 353,000 trending video records collected from multiple countries and includes key metrics such as views, likes, comments, engagement rate, publishing time, and video categories.

The project involved data cleaning and preprocessing in Excel, database creation and analysis using MySQL, and dashboard development in Power BI. Additional analytical features such as engagement rate, like ratio, days to trend, title length, and content segmentation were created to support deeper analysis.

More than 30 SQL queries were executed to uncover patterns related to content categories, channel performance, publishing behavior, audience engagement, and regional trends. The findings were visualized through an interactive Power BI dashboard containing multiple analytical views.

The analysis revealed that Music and Entertainment dominate overall viewership, while countries such as Russia and Mexico exhibit higher audience engagement rates. Publishing time and content category were found to significantly influence trending success. Based on these insights, several recommendations were developed to support content creators, marketers, and digital media stakeholders.

METHODOLOGY

Data Collection

The dataset used for this project contains YouTube Trending Videos data collected from ten countries including the United Kingdom, United States, Canada, India, Germany, France, Mexico, South Korea, Russia, and Japan.

The dataset includes information such as video title, channel name, category, views, likes, dislikes, comments, publishing date, trending date, and country information.

Data Cleaning and Preparation

Data preprocessing was performed using Microsoft Excel before importing the dataset into MySQL.

The following cleaning activities were carried out:

- Removed inconsistencies and formatting issues.
- Standardized column names and data types.
- Validated country codes and category identifiers.
- Removed invalid and unnecessary records.
- Prepared category mapping data for analysis.
- Structured the dataset for SQL processing.

Database Creation

The cleaned dataset was imported into MySQL where a structured database was created.

Separate tables were maintained for video information and category mapping. SQL joins were used to combine category names with the main dataset whenever required.

Feature Engineering

Several analytical columns were created to improve the quality of insights:

Engagement Rate – Measures audience interaction relative to views.

Engagement Rate = $((\text{Likes} + \text{Comments}) / \text{Views}) \times 100$

Like Ratio – Measures positive audience sentiment based on likes and dislikes.

Like Ratio = $(\text{Likes} / (\text{Likes} + \text{Dislikes})) \times 100$

Days to Trend – Calculates the number of days taken for a video to appear on the trending page.

Title Length – Measures the length of video titles for content strategy analysis.

Views Bucket – Classifies videos into High, Medium, and Low view categories.

Engagement Level – Segments videos based on engagement performance.

SQL Analysis

More than 30 SQL queries were executed to perform exploratory and analytical investigations.

The analysis focused on:

- Top performing videos.
- Category performance.
- Channel rankings.
- Country-wise performance.
- Audience engagement patterns.
- Publishing behavior.
- Trending velocity analysis.

Aggregate functions, JOIN operations, CASE statements, date functions, ranking techniques, and grouping operations were used to generate analytical insights.

```
##SECTION 1
```

```
##DATASET OVERVIEW ANALYSIS
```

```
SELECT COUNT(*) AS total_records  
FROM video_analysis_view;
```

```
SELECT COUNT(DISTINCT country_code) AS total_countries  
FROM video_analysis_view;
```

```
SELECT COUNT(DISTINCT category_name) AS total_categories  
FROM video_analysis_view;
```

```
SELECT COUNT(DISTINCT channel_title) AS total_channels  
FROM video_analysis_view;
```

```

SELECT
    country_code,
    COUNT(*) AS total_videos
FROM video_analysis_view
WHERE country_code IS NOT NULL
GROUP BY country_code
ORDER BY total_videos DESC;

```

total_records	total_countries	total_categories
▶ 353351	▶ 9	▶ 18

total_channels
▶ 29927

country_code	total_videos
US	40949
CA	40881
DE	40839
FR	40723
RU	40148
MX	39930
GB	38916
IN	36826
KR	34121

Key Findings from Queries – 1 to 5:

- Dataset contains 353K+ trending video records.
- Data collected from 9 countries.
- Over 29K unique YouTube channels identified.
- Dataset covers 18 video categories.
- US has the highest trending video count.
- The dataset is large and diverse enough for trend and engagement analysis.

```
## SECTION 2
## QUERY 6: Top Viewed Videos Query
SELECT
    title,
    channel_title,
    MAX(views) AS max_views
FROM video_analysis_view
GROUP BY title, channel_title
ORDER BY max_views DESC
LIMIT 10;
## QUERY 7: Top Categories by Total Views
SELECT
    category_name,
    SUM(views) AS total_views
FROM video_analysis_view
GROUP BY category_name
ORDER BY total_views DESC;
```

QUERY 8: Average Views by Category

```
SELECT
    category_name,
    ROUND(AVG(views),0) AS avg_views
FROM video_analysis_view
GROUP BY category_name
ORDER BY avg_views DESC;
```

QUERY 9: Top Performing Channels by Total Views

```
SELECT
    channel_title,
    SUM(views) AS total_views
FROM video_analysis_view
GROUP BY channel_title
ORDER BY total_views DESC
LIMIT 10;
```

QUERY 10: Videos with Highest Likes

```
SELECT
    title,
    channel_title,
    MAX(likes) AS max_likes
FROM video_analysis_view
GROUP BY title, channel_title
ORDER BY max_likes DESC
LIMIT 10;
```

	title	channel_title	max_views
▶	Nicky Jam x J. Balvin - X (EQUIS) Video Oficial ...	NickyJamTV	424538912
	Te Bote Remix - Casper, Nio García, Darell, Nick...	Flow La Movie	337621571
	Bad Bunny - Amorfoda Video Oficial	Bad Bunny	328860380
	Ozuna x Romeo Santos - El Farsante Remix	Ozuna	288811992
	Childish Gambino - This Is America (Official Video)	ChildishGambinoVEVO	259721696
	Drake - God's Plan	DrakeVEVO	258164991
	Ariana Grande - No Tears Left To Cry	ArianaGrandeVevo	208876887
	Becky G, Natti Natasha - Sin Pijama (Official Video)	BeckyGVEVO	200862743
	YouTube Rewind: The Shape of 2017 #YouTu...	YouTube Spotlight	169884583
	Dura - Daddy Yankee (Video Oficial)	Daddy Yankee	167456025

Result 41 x

	category_name	total_views
▶	Music	254362418223
	Entertainment	102342811971
	Film & Animation	27261870898
	People & Blogs	22972584829
	Comedy	21722829746
	Sports	18318438050
	News & Politics	10187470172
	Howto & Style	9650501785
	Science & Technology	8993582312
	Gaming	7543000292
	Education	2722189780

Result 42 x

	category_name	avg_views
►	Music	6192483
	Movies	1954438
	Film & Animation	1391125
	Science & Technology	1127157
	Entertainment	1002290
	Sports	850201
	Comedy	832739
	Gaming	725079
	Howto & Style	536467
	Pets & Animals	520650
	People & Blogs	461510

Result 43 x

Query 6 – Top Viewed Videos:

- Music videos dominated trending views.
- Viral songs maintained long-lasting popularity.
- Entertainment content attracted massive reach.

Query 7 – Top Categories by Total Views:

- Music led total views by a large margin.
- Entertainment ranked second.
- Educational content had lower reach.

Query 8 – Average Views by Category:

- Music had the highest average views.
- Movies and Film & Animation performed strongly.
- Tech content showed solid audience interest.

channel_title	total_views
ChildishGambinoVEVO	10986637068
Marvel Entertainment	10275901157
NickyJamTV	9479859505
Ozuna	8623329509
ibighit	7866843468
DrakeVEVO	7627699734
Bad Bunny	7124207494
ArianaGrandeVevo	6170482942
jypentertainment	5766013743
Ed Sheeran	5762684305

title	channel_title	max_likes
BTS (?????) 'FAKE LOVE' Official MV	ibighit	5613827
Childish Gambino - This Is America (Official Video)	ChildishGambinoVEVO	5444541
Drake - God's Plan	DrakeVEVO	4737873
Bad Bunny - Amorfoda Video Oficial	Bad Bunny	3823879
Ariana Grande - No Tears Left To Cry	ArianaGrandeVevo	3394437
YouTube Rewind: The Shape of 2017 #YouTu...	YouTube Spotlight	3312868
BTS (?????) 'MIC Drop (Steve Aoki Remix)' Offici...	ibighit	2960250
Nicky Jam x J. Balvin - X (EQUIS) Video Oficial ...	NickyJamTV	2818771
Marvel Studios' Avengers: Infinity War Official T...	Marvel Entertainment	2701353
BTS (?????) LOVE YOURSELF ? Tear 'Singularity' ...	ibighit	2700801

Query 9 – Top Performing Channels

- Music channels dominated overall visibility.
- Artist and VEVO channels gained billions of views.
- Major entertainment brands maintained strong reach.

Query 10 – Videos with Highest Likes

- BTS and music artists received the most likes.
- Fan-driven content generated high engagement.
- Music videos outperformed most other categories.

```

## SECTION 3 - ENGAGEMENT ANALYSIS

## Query 11: Top Videos by Engagement Rate
SELECT
    title,
    channel_title,
    ROUND(engagement_rate * 100,2) AS engagement_percentage
FROM video_analysis_view
ORDER BY engagement_rate DESC
LIMIT 10;

## Query 12: Average Engagement Rate by Category
SELECT
    category_name,
    ROUND(AVG(engagement_rate) * 100,2) AS avg_engagement
FROM video_analysis_view
GROUP BY category_name
ORDER BY avg_engagement DESC;

## Query 13: Engagement Level Distribution
SELECT
    CASE
        WHEN engagement_rate >= 0.10 THEN 'High'
        WHEN engagement_rate >= 0.05 THEN 'Medium'
        ELSE 'Low'
    END AS engagement_level,
    COUNT(*) AS total_videos
FROM video_analysis_view
GROUP BY engagement_level
ORDER BY total_videos DESC;

```


title	channel_title	engagement_percentage
Mora Saiyaan Mo Se Bole Na I Shafqat Amanat ...	Mausiqi Studios	95.73
7-7700k / GTX1080ti? 330???? ? ? ? ? ? ? ? ? ? ?	??? TV	87.97
90? ?	??? TV	86.36
?????? ?	Luckyman	82.68
OnePlus 6 Top Features and GIVEAWAY ??- On...	Technical Guruji	77.43
OnePlus 6 Top Features and GIVEAWAY ??- On...	Technical Guruji	77.43
???????? ?	????? TV	74.61
? Te Regalo 10 iPhone X Sorteo Internacional ...	DebRyanShow	72.79

category_name	avg_engagement
Howto & Style	6.21
Gaming	5.91
Education	5.78
Comedy	5.77
Science & Technology	5.76
Pets & Animals	5.3
Music	5.17
Autos & Vehicles	5.09

title	channel_title	max_comments
So Sorry.	Logan Paul Vlogs	1626501
BTS (?????) 'FAKE LOVE' Official MV	ibighit	1228655
YouTube Rewind: The Shape of 2017 #YouTu...	YouTube Spotlight	845233
Suicide: Be Here Tomorrow.	Logan Paul Vlogs	704824
BTS (?????) 'MIC Drop (Steve Aoki Remix)' Offi...	ibighit	559197
Childish Gambino - This Is America (Official Video)	ChildishGambinoVEVO	553371
????????? Samsung Galaxy S9+	Wylsacom	523855
j-hope 'Daydream (???)' MV	ibighit	477233
CHUNKY RICH BROWN LOWDOWN M7D L...	CHUNKY RICH BROWN Official	445261

Query 11 – Top Videos by Engagement Rate

- Tech and regional videos showed very high engagement.
- Strong audience loyalty drove interactions.

Query 12 – Average Engagement Rate by Category

- Howto & Style had the highest engagement.
- Gaming and Education performed strongly.

Query 13 – Engagement Level Distribution

- Most videos had low engagement.
- Only a small portion achieved high interaction.

```

## Query 14: Most Commented Videos
SELECT
    title,
    channel_title,
    MAX(comment_count) AS max_comments
FROM video_analysis_view
GROUP BY title, channel_title
ORDER BY max_comments DESC
LIMIT 10;

## Query 15: Most Liked Categories
SELECT
    category_name,
    SUM(likes) AS total_likes
FROM video_analysis_view
GROUP BY category_name
ORDER BY total_likes DESC;

```

engagement_level	total_videos
Low	242505
Medium	76519
High	34327

category_name	total_likes
Music	7130485003
Entertainment	2808736152
Comedy	1071131798
People & Blogs	681478105
Film & Animation	582846201
Sports	391899997
Howto & Style	344306428
Gaming	294712137

Query 14 – Most Commented Videos

- Music and viral videos received the most comments.
- BTS and Logan Paul generated massive discussions.

Query 15 – Most Liked Categories

- Music dominated total likes.
- Entertainment and Comedy followed behind.

SECTION 4 - TRENDING PATTERN ANALYSIS

Query 16: Trending Videos by Day of Week

```
SELECT
    publish_day,
    COUNT(*) AS total_videos
FROM video_analysis_view
GROUP BY publish_day
ORDER BY total_videos DESC;
```

Query 17: Best Publishing Hour

```
SELECT
    HOUR(publish_hour) AS publish_hour,
    COUNT(*) AS total_videos
FROM video_analysis_view
GROUP BY publish_hour
ORDER BY total_videos DESC;
```

Query 18: Average Days to Trend by Category

```
SELECT
    category_name,
    ROUND(AVG(days_to_trend),2) AS avg_days_to_trend
FROM video_analysis_view
GROUP BY category_name
ORDER BY avg_days_to_trend;
```

publish_hour	total_videos	category_name	avg_days_to_trend
21	1388	Nonprofits & Activism	1.90
10	1308	Shows	1.98
22	1272	Movies	2.39
21	1162	News & Politics	3.23
21	1117	Howto & Style	4.24
22	1103	Entertainment	4.76
20	1094	Gaming	5.34
20	1050	Trailers	5.60

publish_day	total_videos
Fri	58668
Thu	54464
Mon	50832
Wed	50572
Tue	50107
Sat	44913
Sun	43795

Query 19: Fastest Trending Videos

```
SELECT
    title,
    channel_title,
    days_to_trend
FROM video_analysis_view
WHERE days_to_trend = 0
LIMIT 10;
```

Query 20: Slowest Trending Categories

```
SELECT
    category_name,
    ROUND(AVG(days_to_trend),2) AS avg_days
FROM video_analysis_view
GROUP BY category_name
ORDER BY avg_days DESC;
```

category_name	avg_days	title	channel_title	days_to_trend
Education	19.03	2 Weeks with iPhone X	iJustine	0
Music	16.94	The New SpotMini	BostonDynamics	0
Film & Animation	12.20	Slow Mo Katana Sword - 4K - The Slow Mo Guys	The Slow Mo Guys	0
Travel & Events	11.03	Watch live: Sessions testifies to the House Judi...	Washington Post	0
Comedy	9.67	Spilling Tea About Each Other!!	Dolan Twins	0
Science & Technology	8.53	Is the Morphe 3502 Palette Worth the Hype? ...	Jackie Aina	0
People & Blogs	7.11	American Crime Story Season 2: The Assassinat...	TV Promos	0
Autos & Vehides	6.75	The Making of a Song: "Delicate" – watch more ...	Taylor Swift	0

Query 16 – Trending Videos by Day of Week

- Friday had the highest trending activity.
- Weekend uploads showed comparatively lower performance.
- Weekday publishing was more effective overall.

Query 17 – Best Publishing Hour

- Most trending videos were published during evening hours.
- Peak activity was observed around 8 PM to 10 PM.

Query 18 – Average Days to Trend by Category

- Nonprofits & Activism and News categories trended faster.
- Gaming and Entertainment required slightly more time.

Query 19 – Fastest Trending Videos

- Celebrity and tech-related videos trended immediately.
- Strong audience interest accelerated visibility.

Query 20 – Slowest Trending Categories

- Education took the longest time to trend.
- Music and Film categories also needed sustained traction.

Query 23: Highest Engagement Country

```
SELECT
    country_code,
    ROUND(AVG(engagement_rate) * 100,2) AS avg_engagement
FROM video_analysis_view
GROUP BY country_code
ORDER BY avg_engagement DESC;
```

Query 22: Average Views by Country

```
SELECT
    country_code,
    ROUND(AVG(views),0) AS avg_views
FROM video_analysis_view
GROUP BY country_code
ORDER BY avg_views DESC;
```

country_code	avg_engagement
RU	5.79
MX	5.54
FR	5.03
DE	4.58
CA	3.9
US	3.89
GB	3.8
KR	2.53

country_code	avg_views
GB	5911944
US	2360785
CA	1147036
IN	1054712
DE	603462
KR	427163
FR	419924
MX	340129

Query 21: Total Videos by Country

```
SELECT
    country_code,
    COUNT(*) AS total_videos
FROM video_analysis_view
GROUP BY country_code
ORDER BY total_videos DESC;
```

country_code	total_videos
US	40949
CA	40881
DE	40839
FR	40723
RU	40148
MX	39930
GB	38916
IN	36826

Query 24: Most Popular Category by Country

```
SELECT
    country_code,
    category_name,
    COUNT(*) AS total_videos
FROM video_analysis_view
GROUP BY country_code, category_name
ORDER BY total_videos DESC;
```

Query 25: Country with Highest Likes

```
SELECT
    country_code,
    SUM(likes) AS total_likes
FROM video_analysis_view
GROUP BY country_code
ORDER BY total_likes DESC;
```

country_code	total_likes
GB	5234962944
US	3041147198
CA	1618179878
IN	994015552
DE	893392305
FR	708140857
MX	633672370
RU	501579176

country_code	category_name	total_videos
IN	Entertainment	16449
DE	Entertainment	15291
GB	Music	13754
CA	Entertainment	13451
MX	Entertainment	13302
RU	People & Blogs	10206
US	Entertainment	9964
FR	Entertainment	9819

Query 21 – Total Videos by Country

- US uploaded the highest number of trending videos.
- Canada and Germany showed similar trending activity.
- Trending content was concentrated across major global markets.

Query 22 – Average Views by Country

- UK achieved the highest average views per video.
- US and Canada maintained strong audience reach.
- Indian content also showed high average visibility.

Query 23 – Highest Engagement Country

- Russia recorded the strongest engagement rate.
- Mexico and France also showed active audience interaction.
- High engagement was not always linked to highest views.

Query 24 – Most Popular Category by Country

- Entertainment dominated in most countries.
- Music was especially popular in the UK.
- Russia showed stronger preference for People & Blogs content.

Query 25 – Country with Highest Likes

- UK generated the highest total likes.
- US ranked second in audience appreciation.
- Music-heavy regions showed stronger interaction levels.

Query 27: Title Category Distribution

```
SELECT
    title_category,
    COUNT(*) AS total_videosz
FROM video_analysis_view
GROUP BY title_category
ORDER BY total_videos DESC;
```

Query 28: Tag Category Distribution

```
SELECT
    tag_category,
    COUNT(*) AS total_videos
FROM video_analysis_view
WHERE tag_category IN ('High', 'Medium', 'Low')
GROUP BY tag_category
ORDER BY total_videos DESC;
```

Query 26: Average Title Length by Category

```
SELECT
    category_name,
    ROUND(AVG(title_length),0) AS avg_title_length
FROM video_analysis_view
GROUP BY category_name
ORDER BY avg_title_length DESC;
```

category_name	avg_title_length
Movies	81
Sports	64
News & Politics	60
Travel & Events	60
Entertainment	56
Education	53
Howto & Style	53
People & Blogs	52
Music	52
Autos & Vehicles	52

title_category	total_videos
Medium	162187
Short	105104
Long	86048
High	6
Low	4

	tag_category	total_videos
►	High	177372
	Medium	105856
	Low	70105

Query 30: Channels with Highest Average Engagement

```
SELECT
    channel_title,
    ROUND(AVG(engagement_rate) * 100,2) AS avg_engagement
FROM video_analysis_view
GROUP BY channel_title
HAVING COUNT(*) >= 5
ORDER BY avg_engagement DESC
LIMIT 10;
```

Query 29: Average Tag Count by Category

```
SELECT
    category_name,
    ROUND(AVG(tag_count),0) AS avg_tag_count
FROM video_analysis_view
GROUP BY category_name
ORDER BY avg_tag_count DESC;
```


category_name	avg_tag_count		channel_title	avg_engagement
Pets & Animals	22		Oka	37.75
Movies	22	▶	koreanvibe	36.93
Howto & Style	21		Diego Granados	36.65
Travel & Events	21		ElvisaYomastercard	29.33
Comedy	20		Cornelio Vega	28.83
Education	20		SUKOJIIVSURAM	28.7
Entertainment	19		Zeo	27.94

Tag and title Insights:

- Movie, Sports, and News videos used longer titles, likely to provide more context and attract clicks.
- Medium-length titles performed best overall, suggesting balanced and readable titles gain more traction.
- Videos with higher tag usage appeared more often in trending results, indicating better discoverability.
- Pets & Animals, Movies, and Travel categories relied heavily on tags to improve search visibility.
- Highly engaged channels were not always the biggest channels, showing that audience interaction matters more than size alone.

Dashboard Development

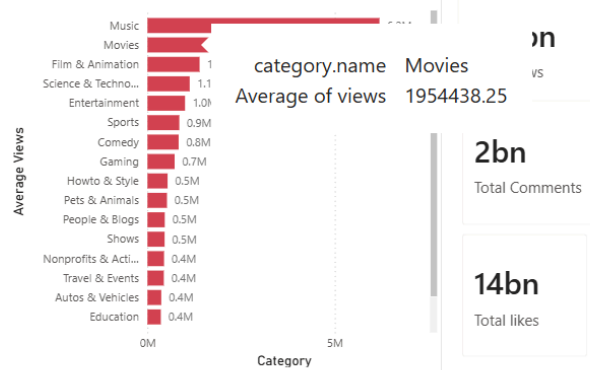
Power BI was used to develop a six-page interactive dashboard.

The dashboard includes:

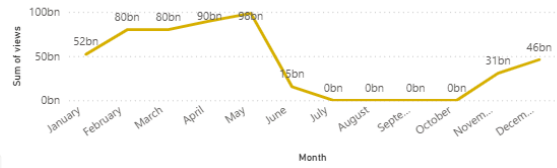
- KPI cards.
- Interactive slicers.
- Drill-through capabilities.
- Cross-filtering visuals.
- Category analysis.
- Channel analysis.
- Engagement analysis.
- Executive summary dashboard.

Youtube Analytics - Final Project

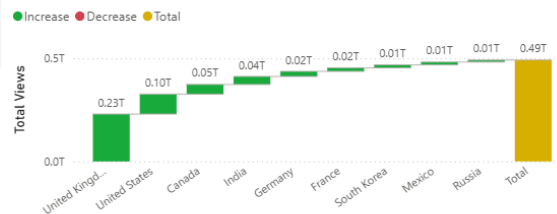
Average of views by category.name



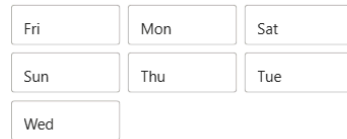
Views Trend Over Time



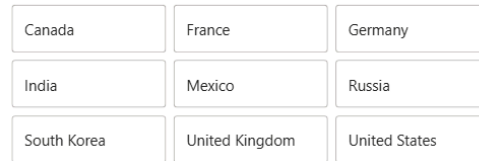
Views by country



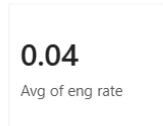
Publish day



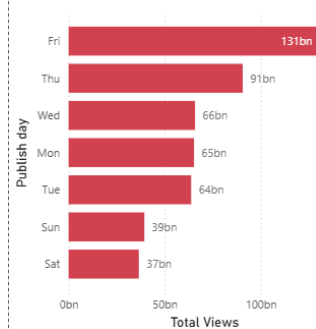
Country code



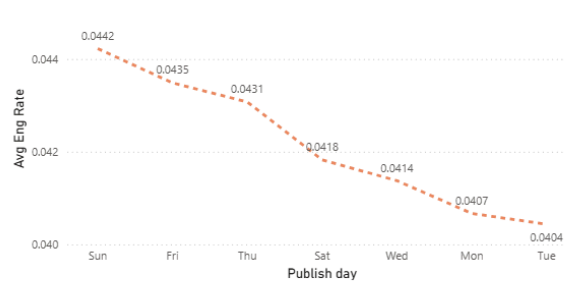
Country



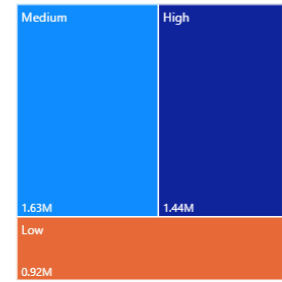
Views by Publish Day



Average of eng rate by publish day

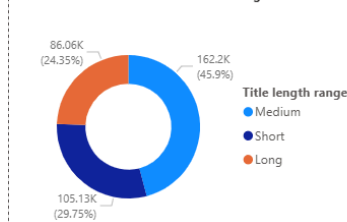


Average of views by Tag category

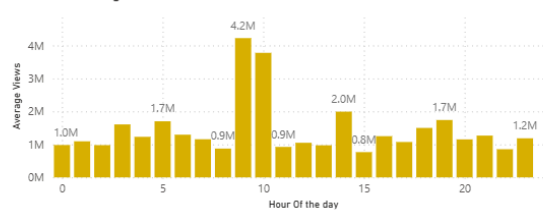


Content Strategy Analysis

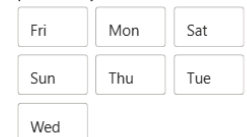
No of Videos with various Title length



Best Publishing Hour

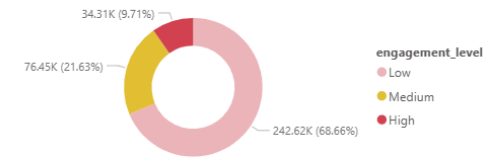


publish day



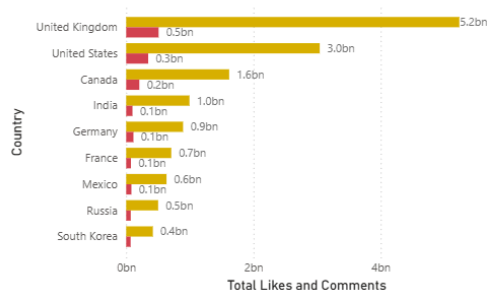
Engagement Analysis

Engagement Level Distribution

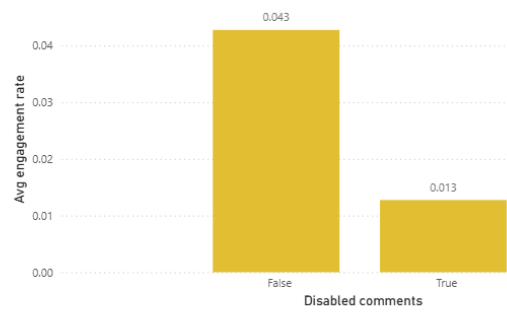


Likes vs Comments by Country

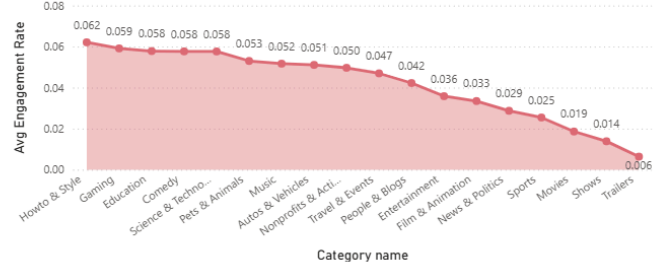
Sum of likes Sum of comment_count



Impact of Comment Availability on Engagement

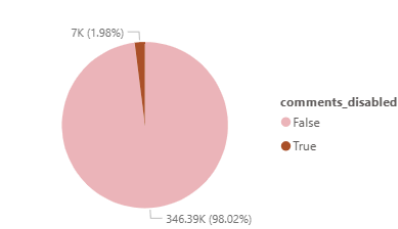


Category Engagement Ranking

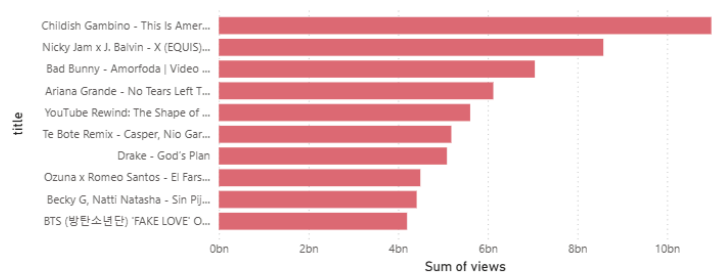


Video Performance Analysis

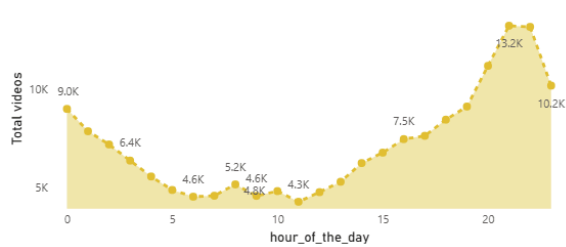
Comment Availability Distribution



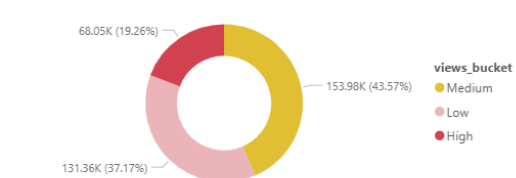
Top 10 Trending Videos by Views

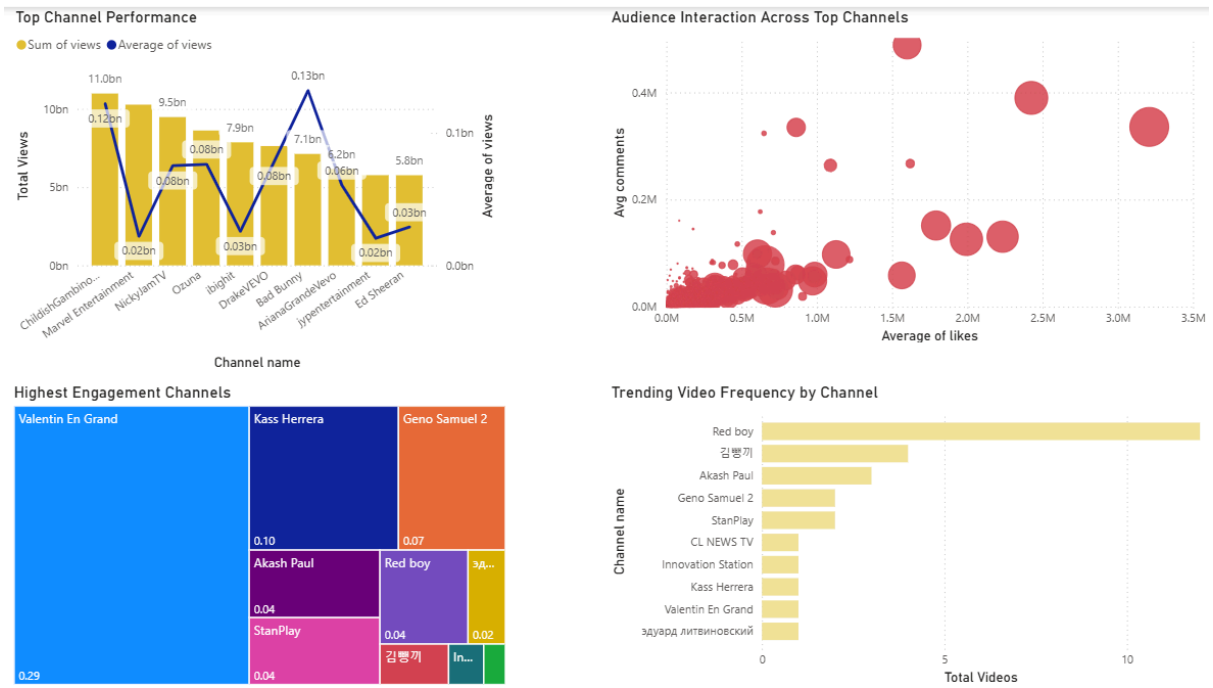


Trending Videos by Publishing Hour

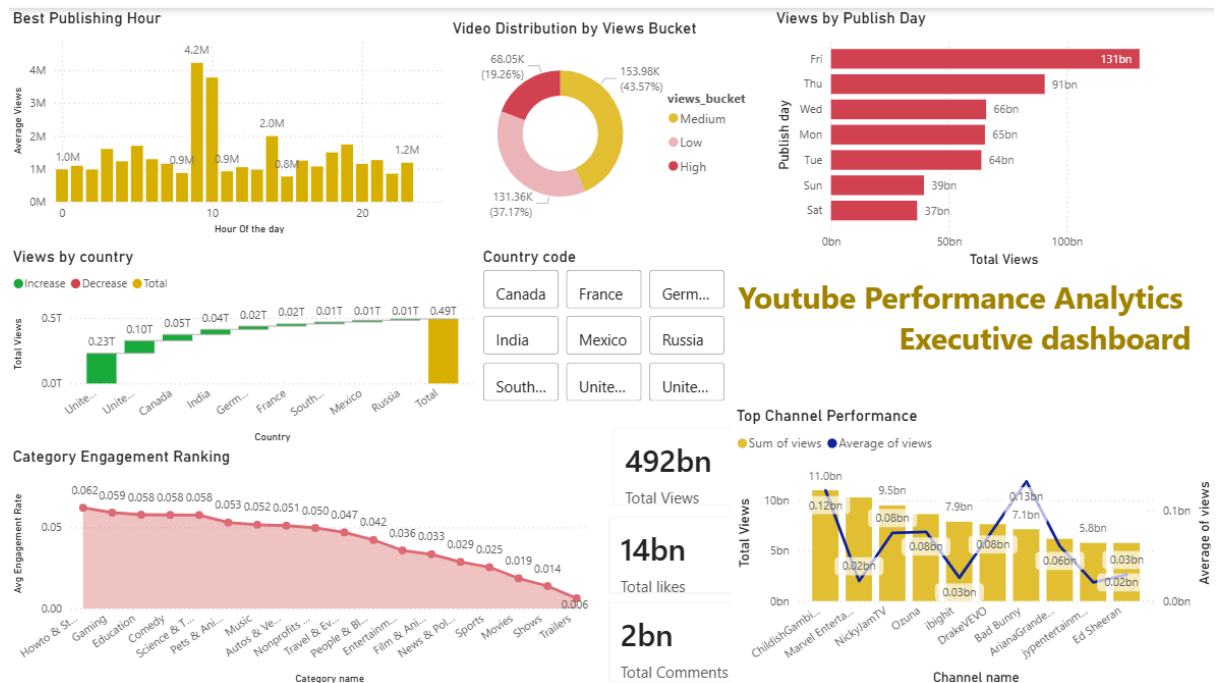


Video Distribution by View Range





OVERALL DASHBOARD:



ANALYSIS FINDINGS

Category Analysis

Entertainment was the most common category among trending videos, indicating its strong presence on the platform. However, Music generated the highest total views and dominated the most-viewed content rankings.

Music videos consistently appeared among the highest-performing content and contributed significantly to overall platform viewership.

Country Analysis

The United Kingdom recorded the highest average views per trending video among all analyzed countries, The United States and Canada also contributed substantial viewership.

Russia and Mexico recorded the highest average engagement rates, indicating stronger audience interaction compared to other regions.

Channel Analysis

Several channels demonstrated exceptional performance by consistently generating high view counts and audience interaction.

The analysis showed that top-performing channels maintained both high visibility and strong engagement metrics, highlighting the importance of channel reputation and audience loyalty.

Engagement Analysis

Videos with active comment sections achieved significantly higher engagement rates than videos with comments disabled.

Audience participation through comments appears to play an important role in improving overall engagement performance.

High engagement content was concentrated within specific categories such as How-to & Style, Gaming, Education, and Comedy.

Publishing Behavior Analysis

Publishing time showed a noticeable impact on trending performance.

Videos published during evening hours achieved higher visibility and were more likely to trend successfully.

Friday and Thursday emerged as the strongest publishing days in terms of total views and audience engagement.

Video Performance Analysis

Most trending videos reached the trending page within a short period after publication, indicating that early audience response is critical for success.

The top-performing videos accumulated hundreds of millions of views and significantly outperformed the average trending video.

BUSINESS INSIGHTS

Content category is one of the strongest drivers of performance. Music and Entertainment content consistently attract larger audiences and generate higher view counts.

Publishing strategy significantly influences content success. Videos released during peak audience activity periods have a greater probability of trending.

Audience interaction directly contributes to engagement performance. Videos encouraging discussion and participation achieve stronger results.

Regional audience behavior varies considerably. High-engagement markets provide valuable opportunities for targeted campaigns and audience growth.

Successful channels combine content quality, publishing consistency, and audience engagement to maintain long-term performance.

RECOMMENDATIONS

1. Prioritize High-Performing Categories

Creators should focus on categories such as Music and Entertainment, which consistently generate higher views and wider audience reach.

2. Publish During Peak Hours

Videos should be scheduled between 8 PM and 11 PM to maximize visibility and increase the probability of appearing on trending lists.

3. Encourage Audience Interaction

Creators should actively encourage comments, discussions, polls, challenges, and interactive content formats to improve engagement rates.

4. Optimize Metadata

Titles, tags, and descriptions should be carefully optimized to improve discoverability and search performance.

5. Target High-Engagement Markets

Marketing campaigns should prioritize countries with strong engagement metrics, particularly Russia and Mexico.

6. Leverage Collaborations

Partnerships with established channels can help creators expand audience reach and improve content visibility.

7. Maintain Consistent Publishing Schedules

Regular content publishing helps strengthen audience retention and improves channel performance over time.

8. Monitor Performance Metrics Continuously

Creators should regularly track engagement rate, like ratio, and audience interaction metrics to identify content improvement opportunities.

CONCLUSION

This project successfully transformed a large YouTube Trending Videos dataset into meaningful business insights through the application of Excel, MySQL, SQL analysis, and Power BI visualization techniques.

The study demonstrated that content category, publishing time, audience engagement, and regional behavior are key factors influencing trending success on YouTube. Music and Entertainment emerged as the strongest-performing categories, while engagement-focused content generated deeper audience interaction.

The Power BI dashboard provided an interactive platform for exploring trends across countries, categories, channels, and engagement metrics. The resulting insights and recommendations offer practical guidance for content creators, marketers, and digital media professionals seeking to improve content performance and audience reach.

Overall, the project highlights the value of data analytics in understanding digital content behavior and supporting data-driven decision-making within the online video ecosystem.